

Лабораторная работа №3

Настройка DHCP-сервера

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1. Цели и задачи работы

2. Процесс выполнения лабораторной работы

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Целью данной работы является приобретение практических навыков установки, настройки и проверки работы DHCP-сервера Kea на виртуальной машине **server**.

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- проверить DNS-запись клиента с помощью **dig**;
- подготовить сценарий автоматической настройки DHCP-сервера для Vagrant.

2. 2. Процесс выполнения лабораторной работы

2.1 Устанавливаю DHCP-сервер Kea

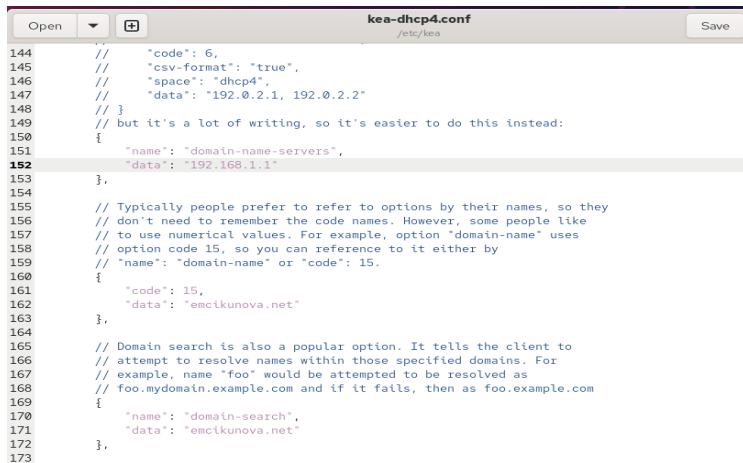
```
-----
Total                                                    13 MB/s | 5.2 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : mariadb-connector-c-config-3.4.4-2.el10_2.noarch 1/6
  Installing     : mariadb-connector-c-3.4.4-2.el10_2.x86_64      2/6
  Installing     : log4cplus-2.1.1-10.el10.x86_64                 3/6
  Installing     : libpq-16.14-1.el10_2.x86_64                    4/6
  Installing     : kea-libs-3.0.3-1.el10_2.x86_64                 5/6
Running scriptlet: kea-3.0.3-1.el10_2.x86_64                     6/6
  Installing     : kea-3.0.3-1.el10_2.x86_64                     6/6
Running scriptlet: kea-3.0.3-1.el10_2.x86_64                     6/6

Installed:
  kea-3.0.3-1.el10_2.x86_64           kea-libs-3.0.3-1.el10_2.x86_64
  libpq-16.14-1.el10_2.x86_64        log4cplus-2.1.1-10.el10.x86_64
  mariadb-connector-c-3.4.4-2.el10_2.x86_64 mariadb-connector-c-config-3.4.4-2.el10_2.noarch

Complete!
[root@server.emcikunova.net ~]#
```

Рисунок 1: Установка DHCP-сервера Kea

2.2 Настраиваю DNS-параметры DHCP



```
144 //      "code": 6,
145 //      "csv-format": "true",
146 //      "space": "dhcp4",
147 //      "data": "192.0.2.1, 192.0.2.2"
148 // }
149 // but it's a lot of writing, so it's easier to do this instead:
150 {
151     "name": "domain-name-servers",
152     "data": "192.168.1.1"
153 },
154
155 // Typically people prefer to refer to options by their names, so they
156 // don't need to remember the code names. However, some people like
157 // to use numerical values. For example, option "domain-name" uses
158 // option code 15, so you can reference to it either by
159 // "name": "domain-name" or "code": 15.
160 {
161     "code": 15,
162     "data": "emcikunova.net"
163 },
164
165 // Domain search is also a popular option. It tells the client to
166 // attempt to resolve names within those specified domains. For
167 // example, name "foo" would be attempted to be resolved as
168 // foo.mydomain.example.com and if it fails, then as foo.example.com
169 {
170     "name": "domain-search",
171     "data": "emcikunova.net"
172 },
173
```

Рисунок 2: Настройка DNS-сервера и домена в kea-dhcp4.conf

2.3 Задаю DHCP-подсеть и пул адресов

```
288 // parameters. One of those parameters is "pools" that is also a list of
289 // structures.
290 "subnet4": [
291     {
292
293         "id": 1,
294         "subnet": "192.168.1.0/24",
295
296         "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],
297         "option-data": [
298             {
299                 "name": "routers",
300                 "data": "192.168.1.1"
301             }
302         ]
303         // You can add more subnets there.
304     }
305 ],
306
```

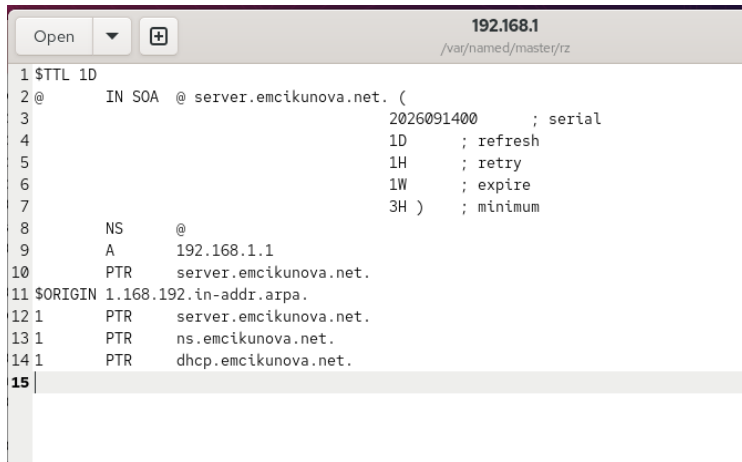
Рисунок 3: Настройка сети 192.168.1.0/24 и диапазона адресов

2.4 Привязываю DHCP к интерфейсу eth1

```
27 // and will be ignored by other components.
28 "Dhcp4": {
29     // Add names of your network interfaces to listen on.
30     "interfaces-config": {
31         // See section 8.2.4 for more details. You probably want to add just
32         // interface name (e.g. "eth0" or specific IPv4 address on that
33         // interface name (e.g. "eth0/192.0.2.1").
34         "interfaces": [ "eth1" ]
35
36         // Kea DHCPv4 server by default listens using raw sockets. This ensures
37         // all packets, including those sent by directly connected clients
38         // that don't have IPv4 address yet, are received. However, if your
39         // traffic is always relayed, it is often better to use regular
40         // UDP sockets. If you want to do that, uncomment this line:
41         // "dhcp-socket-type": "udp"
42     },
43
44     // Kea supports control channel which is a way to receive management
```

Рисунок 4: Настройка interfaces-config для eth1

2.5 Добавляю DHCP-сервер в обратную DNS-зону



```
1 $TTL 1D
2 @      IN SOA  @ server.emcikunova.net. (
3                               2026091400      ; serial
4                               1D              ; refresh
5                               1H              ; retry
6                               1W              ; expire
7                               3H )            ; minimum
8      NS      @
9      A      192.168.1.1
10     PTR     server.emcikunova.net.
11 $ORIGIN 1.168.192.in-addr.arpa.
12 1       PTR     server.emcikunova.net.
13 1       PTR     ns.emcikunova.net.
14 1       PTR     dhcp.emcikunova.net.
15
```

Рисунок 5: Добавление PTR-записи dhcp.emcikunova.net

2.6 Проверяю доступность DHCP-сервера по имени

```
[root@server.emcikunova.net ~]# gedit /var/named/master/fz/emcikunova.net
[root@server.emcikunova.net ~]# gedit /var/named/master/rz/192.168.1
[root@server.emcikunova.net ~]#
[root@server.emcikunova.net ~]# systemctl restart named
[root@server.emcikunova.net ~]# ping dhcp.emcikunova.net
PING dhcp.emcikunova.net (192.168.1.1) 56(84) bytes of data.
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.015 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.111 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.085 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=5 ttl=64 time=0.084 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=6 ttl=64 time=0.088 ms
64 bytes from ns.emcikunova.net (192.168.1.1): icmp_seq=7 ttl=64 time=0.112 ms
^C
--- dhcp.emcikunova.net ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6199ms
rtt min/avg/max/mdev = 0.015/0.078/0.112/0.031 ms
[root@server.emcikunova.net ~]#
```

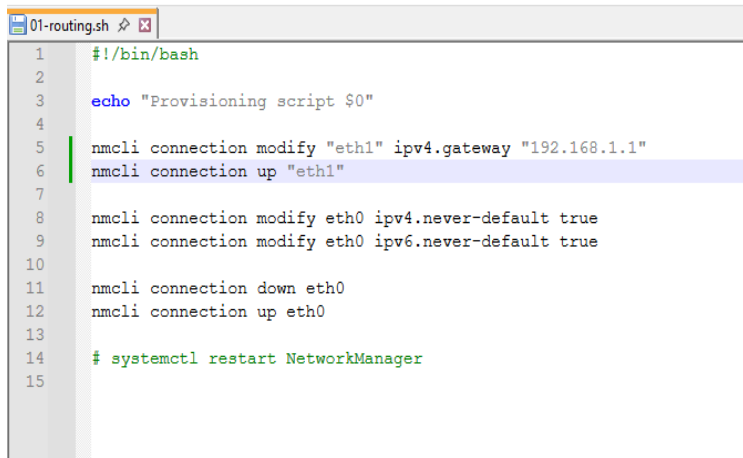
Рисунок 6: Проверка dhcp.emcikunova.net с помощью ping

2.7 Настраиваю firewalld и запускаю DHCP

```
[root@server.emcikunova.net ~]#  
[root@server.emcikunova.net ~]# firewall-cmd --add-service=dhcp  
success  
[root@server.emcikunova.net ~]# firewall-cmd --add-service=dhcp --permanent  
success  
[root@server.emcikunova.net ~]# restorecon -vR /etc  
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0  
[root@server.emcikunova.net ~]# restorecon -vR /var/named/  
[root@server.emcikunova.net ~]# restorecon -vR /var/lib/kea/  
[root@server.emcikunova.net ~]# systemctl start kea-dhcp4.service  
[root@server.emcikunova.net ~]#
```

Рисунок 7: Разрешение DHCP в firewalld и запуск kea-dhcp4

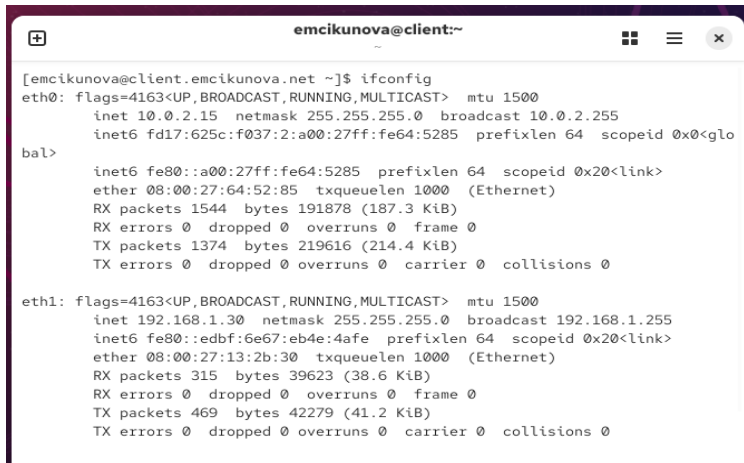
2.8 Создаю скрипт маршрутизации клиента



```
01-routing.sh
1  #!/bin/bash
2
3  echo "Provisioning script $0"
4
5  nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
6  nmcli connection up "eth1"
7
8  nmcli connection modify eth0 ipv4.never-default true
9  nmcli connection modify eth0 ipv6.never-default true
10
11 nmcli connection down eth0
12 nmcli connection up eth0
13
14 # systemctl restart NetworkManager
15
```

Рисунок 8: Скрипт 01-routing.sh для виртуальной машины client

2.9 Проверяю сетевые интерфейсы клиента



```
emcikunova@client:~  
[emcikunova@client.emcikunova.net ~]$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255  
    inet6 fd17:625c:f037:2:a00:27ff:fe64:5285  prefixlen 64  scopeid 0x0<global>  
    ether 08:00:27:64:52:85  txqueuelen 1000  (Ethernet)  
    RX packets 1544  bytes 191878 (187.3 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 1374  bytes 219616 (214.4 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 192.168.1.30  netmask 255.255.255.0  broadcast 192.168.1.255  
    inet6 fe80::edbf:6e67:eb4e:4afe  prefixlen 64  scopeid 0x20<link>  
    ether 08:00:27:13:2b:30  txqueuelen 1000  (Ethernet)  
    RX packets 315  bytes 39623 (38.6 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 469  bytes 42279 (41.2 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

Рисунок 9: Вывод ifconfig на виртуальной машине client

2.10 Анализирую файл DHCP-аренд

```
[root@server.emcikunova.net ~]# cat /var/lib/kea/kea-leases4.csv  
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id  
192.168.1.30,08:00:27:13:2b:30,01:08:00:27:13:2b:30,3600,1789388205,1,0,0,client,0,,0  
[root@server.emcikunova.net ~]#
```

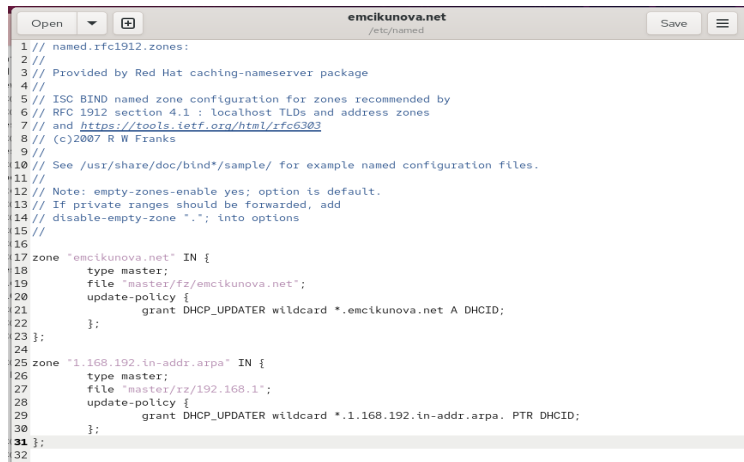
Рисунок 10: Содержимое файла kea-leases4.csv

2.11 Создаю TSIG-ключ для DDNS

```
[root@server.emcikunova.net ~]#  
[root@server.emcikunova.net ~]# mkdir -p /etc/named/keys  
[root@server.emcikunova.net ~]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key  
[root@server.emcikunova.net ~]# cat /etc/named/keys/dhcp_updater.key  
key "DHCP_UPDATER" {  
    algorithm hmac-sha512;  
    secret "xl04u02CuQ4Np2mymhoideoqzzTJTm5RjS9W2zMGH8TF0uNNhbaa+H+tgGae0c320PYJa5M7PrBX16nbJC3Q7A==";  
};  
[root@server.emcikunova.net ~]# chown -R named:named /etc/named/keys/  
[root@server.emcikunova.net ~]# gedit /etc/named.conf  
[root@server.emcikunova.net ~]#
```

Рисунок 11: Создание ключа DHCP_UPDATER

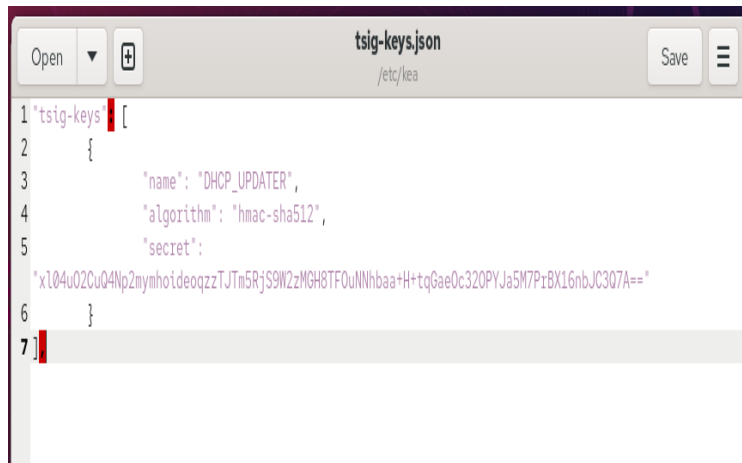
2.12 Разрешаю обновление DNS-зон



```
1 // named.rfc1912.zones:
2 //
3 // Provided by Red Hat caching-nameserver package
4 //
5 // ISC BIND named zone configuration for zones recommended by
6 // RFC 1912 section 4.1 : localhost TLDs and address zones
7 // and https://tools.ietf.org/html/rfc6303
8 // (c)2007 R W Franks
9 //
10 // See /usr/share/doc/bind*/sample/ for example named configuration files.
11 //
12 // Note: empty-zones-enable yes; option is default.
13 // If private ranges should be forwarded, add
14 // disable-empty-zone "."; into options
15 //
16
17 zone "emicikunova.net" IN {
18     type master;
19     file "master/fz/emicikunova.net";
20     update-policy {
21         grant DHCP_UPDATER wildcard *.emicikunova.net A DHCPID;
22     };
23 };
24
25 zone "1.168.192.in-addr.arpa" IN {
26     type master;
27     file "master/rz/192.168.1";
28     update-policy {
29         grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa. PTR DHCPID;
30     };
31 };
32
```

Рисунок 12: Настройка update-policy для прямой и обратной зон

2.13 Создаю ключ Кеа в формате JSON



```
1 "tsig-keys" : [  
2     {  
3         "name": "DHCP_UPDATER",  
4         "algorithm": "hmac-sha512",  
5         "secret":  
6         "x104u02Cu04Np2mymhoideoqzTJTm5RjS9W2zMGH8TF0uNNhbaa+H+tqGae0c320PYJa5M7PrBX16nbJC3Q7A=="  
7     }  
8 ],
```

Рисунок 13: Файл tsig-keys.json

2.14 Настраиваю Kea DHCP-DDNS



```
17 // DHCP-DDNS configuration starts here. This is a very simple configuration
18 // that simply starts the DDNS daemon, but will not do anything useful.
19 // See Section 11 for examples and details description.
20 "DhcpDdns":
21 {
22   "ip-address": "127.0.0.1",
23   "port": 53001,
24   "control-socket": {
25     "socket-type": "unix",
26     "socket-name": "kea-ddns-ctrl-socket"
27   },
28   <?include "/etc/kea/tsig-keys.json"?>
29   "forward-ddns" : {
30     "ddns-domains" : [
31       {
32         "name": "emcikunova.net.",
33         "key-name": "DHCP_UPDATER",
34         "dns-servers": [
35           { "ip-address": "192.168.1.1" }
36         ]
37       }
38     ]
39   },
40   "reverse-ddns" : {
41     "ddns-domains" : [
42       {
43         "name": "1.168.192.in-addr.arpa.",
44         "key-name": "DHCP_UPDATER",
45         "dns-servers": [
46           { "ip-address": "192.168.1.1" }
47         ]
48       }
49     ]
50   }
51 },
52 }
```

Рисунок 14: Файл kea-dhcp-ddns.conf

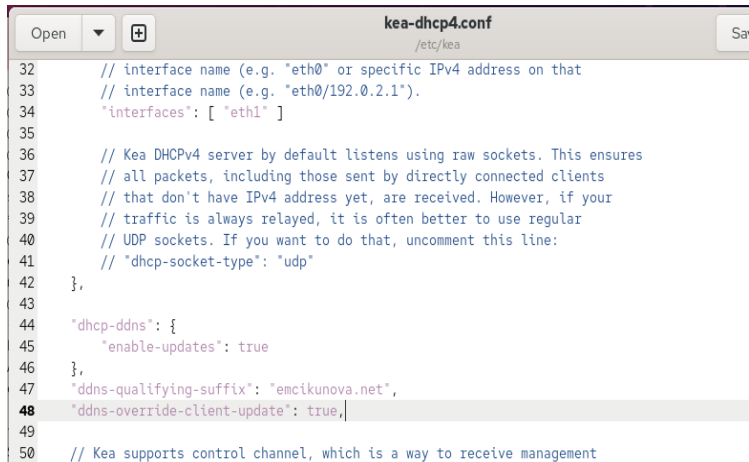
2.15 Проверяю и запускаю службу DHCP-DDNS

```
[root@server.emcikunova.net ~]#
[root@server.emcikunova.net ~]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server.emcikunova.net ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2026-09-14 11:39:28.704 INFO [kea-dhcp-ddns.dctl/17924.140627666033984] DCTL_CONFIG_CHECK_COMPLETE serv
er has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0),
text=Configuration check successful
[root@server.emcikunova.net ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service' -> '/usr/lib/systemd/s
ystem/kea-dhcp-ddns.service'.
[root@server.emcikunova.net ~]# systemctl status kea-dhcp-ddns.service
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-14 11:39:57 UTC; 11s ago
 Invocation: 19cdbbf8497d4732a168cdb8b03e1e98
    Docs: man:kea-dhcp-ddns(8)
   Main PID: 18177 (kea-dhcp-ddns)
    Status: "Dispatching packets..."
     Tasks: 5 (limit: 10233)
    Memory: 2.3M (peak: 7M)
       CPU: 12ms
    CGroup: /system.slice/kea-dhcp-ddns.service
           └─18177 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 14 11:39:57 server.emcikunova.net systemd[1]: Starting kea-dhcp-ddns.service - Kea DHCP-DDNS Server>
Sep 14 11:39:57 server.emcikunova.net kea-dhcp-ddns[18177]: 2026-09-14 11:39:57.262 INFO [kea-dhcp-ddn>
Sep 14 11:39:57 server.emcikunova.net kea-dhcp-ddns[18177]: INFO COMMAND_ACCEPTOR_START Starting to ac>
Sep 14 11:39:57 server.emcikunova.net kea-dhcp-ddns[18177]: INFO DCTL_CONFIG_COMPLETE server has compl>
Sep 14 11:39:57 server.emcikunova.net kea-dhcp-ddns[18177]: INFO DHCP_DDNS_STARTED Kea DHCP-DDNS serve>
Sep 14 11:39:57 server.emcikunova.net systemd[1]: Started kea-dhcp-ddns.service - Kea DHCP-DDNS Server.
[root@server.emcikunova.net ~]#
```

Рисунок 15: Проверка конфигурации и запуск kea-dhcp-ddns

2.16 Включаю DDNS в DHCPv4



```
32 // interface name (e.g. "eth0" or specific IPv4 address on that
33 // interface name (e.g. "eth0/192.0.2.1").
34 "interfaces": [ "eth1" ]
35
36 // Kea DHCPv4 server by default listens using raw sockets. This ensures
37 // all packets, including those sent by directly connected clients
38 // that don't have IPv4 address yet, are received. However, if your
39 // traffic is always relayed, it is often better to use regular
40 // UDP sockets. If you want to do that, uncomment this line:
41 // "dhcp-socket-type": "udp"
42 },
43
44 "dhcp-ddns": {
45     "enable-updates": true
46 },
47 "ddns-qualifying-suffix": "emcikunova.net",
48 "ddns-override-client-update": true,
49
50 // Kea supports control channel, which is a way to receive management
```

Рисунок 16: Добавление dhcp-ddns в kea-dhcp4.conf

2.17 Перезапускаю DHCPv4-сервер

```
[root@server.emcikunova.net ~]#  
[root@server.emcikunova.net ~]# systemctl restart kea-dhcp4.service  
[root@server.emcikunova.net ~]# systemctl status kea-dhcp4.service  
● kea-dhcp4.service - Kea DHCPv4 Server  
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)  
   Active: active (running) since Mon 2026-09-14 11:42:58 UTC; 11s ago  
 Invocation: 50e59e8cb68c4b0d9b1e5b595e42e7e2  
    Docs: man:kea-dhcp4(8)  
   Main PID: 18695 (kea-dhcp4)  
   Status: "Dispatching packets..."  
   Tasks: 7 (limit: 10233)  
  Memory: 2.5M (peak: 7M)  
     CPU: 13ms  
   CGroup: /system.slice/kea-dhcp4.service  
           └─18695 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf  
  
Sep 14 11:42:58 server.emcikunova.net systemd[1]: Starting kea-dhcp4.service - Kea DHCPv4 Server...  
Sep 14 11:42:58 server.emcikunova.net kea-dhcp4[18695]: 2026-09-14 11:42:58.584 INFO [kea-dhcp4.dhcp4/>  
Sep 14 11:42:58 server.emcikunova.net kea-dhcp4[18695]: 2026-09-14 11:42:58.585 INFO [kea-dhcp4.comman>  
Sep 14 11:42:58 server.emcikunova.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.  
[root@server.emcikunova.net ~]#
```

Рисунок 17: Проверка состояния kea-dhcp4.service

2.18 Проверяю DNS-запись клиента

```
[emcikunova@client.emcikunova.net ~]$ nmcli connection down eth1
Connection 'eth1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
[emcikunova@client.emcikunova.net ~]$ nmcli connection up eth1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
[emcikunova@client.emcikunova.net ~]$ dig @192.168.1.1 client.emcikunova.net

; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.emcikunova.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29917
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 60cbf970d263840f010000006aa7de1c932c4f5dd3cc2792 (good)
;; QUESTION SECTION:
;client.emcikunova.net.          IN      A

;; ANSWER SECTION:
client.emcikunova.net. 1200     IN      A      192.168.1.30

;; Query time: 0 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Mon Sep 14 11:44:27 UTC 2026
;; MSG SIZE rcvd: 94

[emcikunova@client.emcikunova.net ~]$
```

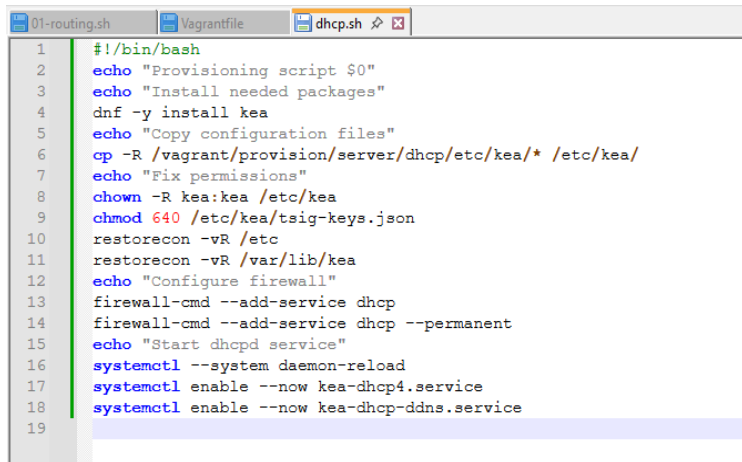
Рисунок 18: Проверка client.emcikunova.net с помощью dig

2.19 Копирую конфигурацию для Vagrant

```
[root@server.emcikunova.net ~]#  
[root@server.emcikunova.net ~]# cd /vagrant/provision/server/  
[root@server.emcikunova.net server]# mkdir -p /vagrant/provision/server/dhcp/etc/kea  
[root@server.emcikunova.net server]# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/  
[root@server.emcikunova.net server]# cd /vagrant/provision/server/dns/  
[root@server.emcikunova.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/  
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/emcikunova.net'? y  
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error  
r  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error  
[root@server.emcikunova.net dns]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/  
cp: overwrite '/vagrant/provision/server/dns/etc/named/emcikunova.net'? y  
[root@server.emcikunova.net dns]# cd /vagrant/provision/server/  
[root@server.emcikunova.net server]# touch dhcp.sh  
[root@server.emcikunova.net server]# chmod +x dhcp.sh  
[root@server.emcikunova.net server]#
```

Рисунок 19: Копирование файлов DHCP и DNS

2.20 Создаю сценарий автоматической настройки



```
01-routing.sh Vagrantfile dhcp.sh
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install kea
5  echo "Copy configuration files"
6  cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
7  echo "Fix permissions"
8  chown -R kea:kea /etc/kea
9  chmod 640 /etc/kea/tsig-keys.json
10 restorecon -vR /etc
11 restorecon -vR /var/lib/kea
12 echo "Configure firewall"
13 firewall-cmd --add-service dhcp
14 firewall-cmd --add-service dhcp --permanent
15 echo "Start dhcpd service"
16 systemctl --system daemon-reload
17 systemctl enable --now kea-dhcp4.service
18 systemctl enable --now kea-dhcp-ddns.service
19
```

Рисунок 20: Скрипт dhcp.sh для настройки DHCP-сервера

2.21 Вывод

В ходе лабораторной работы был установлен и настроен DHCP-сервер Kea на виртуальной машине **server**.

Для внутренней сети **192.168.1.0/24** был задан пул адресов

192.168.1.30–192.168.1.199, а клиентская машина получила адрес **192.168.1.30** через DHCP.

Дополнительно было настроено динамическое обновление DNS-зоны с использованием Kea DHCP-DDNS и TSIG-ключа **DHCP_UPDATER**.

Работа DHCP и DDNS была проверена с помощью утилит **ifconfig**, **ping**, **cat** и **dig**. Также был подготовлен сценарий **dhcp.sh**, автоматизирующий установку и настройку DHCP-сервера во внутреннем окружении Vagrant.